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Ex # 13.1

Date:

1. Let $f(x,y) = x^2y + 1$. Find

a) $f(2,1)$, b) $f(1,2)$, c) $f(0,0)$, d) $f(1,-3)$
e) $f(3a, a)$ f) $f(ab, a-b)$.

a) solve $f(2,1) = (2)^2(1) + 1 = 4+1 = \boxed{5}$
b) $f(1,2) = (1)^2(2) + 1 = 2+1 = 3$
c) $f(0,0) = (0)^2(0) + 1 = 0+1 = 1$
d) $f(1,-3) = (1)^2(-3) + 1 = -3+1 = -2$
e) $f(3a, a) = (3a)^2(a) + 1 = 9a^2(a) + 1 = 9a^3 + 1$
f) $f(ab, a-b) = (ab)^2(a-b) + 1 = a^2b^2(a-b) + 1$

2. Let $f(x,y) = x + \sqrt[3]{xy}$. Find

a) $f(t, t^2)$

solve

$$f(t,t^2) = t + \sqrt[3]{t(t^2)}$$
$$= t + \sqrt[3]{t^3}$$
$$= t + t = 2t$$

b) $f(x, x^2)$

solve

$$f(t,t^2) = t + \sqrt[3]{t(t^2)}$$
$$= x + \sqrt[3]{x^3}$$
$$= x + x = 2x$$

c) $f(2y^2, 4y)$

solve

$$f(2y^2, 4y) = 2y^2 + \sqrt[3]{(2y^2)(4y)}$$
$$= 2y^2 + \sqrt[3]{8y^3}$$
$$= 2y^2 + \sqrt[3]{(2y)^3}$$
$$= 2y^2 + 2y$$

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3. Let $f(x,y) = xy + 3$. Find.

a) $f(x+y, x-y)$

solve

$$f(x+y, x-y) = (x+y)(x-y) + 3$$
$$= x^2 - y^2 + 3$$

b) $f(x, y, 3x^2y^3)$

solve

$$f(xy, 3x^2y^3) = (xy)(3x^2y^3) + 3$$
$$= 3x^3y^4 + 3$$
$$= 3(x^3y^4 + 1)$$

9. Suppose that the concentration C in mg/L of medication in a patient's bloodstream is modeled by the function $C(x,t) = 0.2x(e^{-0.2t} - e^{-t})$, where x is the dosage of medication in mg and t is the number of hours since the beginning of administration of the medication.

a) Estimate the value of $C(25, 3)$ to two decimal places. Include appropriate units and interpret your answer in a physical context.

solve

We are given: Dosage $x = 25$ mg.
Time $t = 3$ hours.

$$C(25,t) = 0.2(25)(e^{-0.2(3)} - e^{-3}) = 5(e^{-0.6} - e^{-3})$$
$$= 2.50 \text{ mg/L}.$$

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b) If the dosage is 100 mg, give a formula for the concentration as a function of time t.

solve

$$C(100, t) = 0.2 \times 100 (e^{-0.2t} - e^{-t})$$
$$c(t) = 20(e^{-0.2t} - e^{-t})$$

c) Give a formula that describes the concentration after 1 hour in term of the dosage x.

solve

$$C(x, 1) = 0.2x(e^{-0.2(1)} - e^{-1})$$
$$C(x) = 0.2x(0.4508)$$
$$C(x) = 0.0902x \quad mg/L \text{ after 1 hour.}$$

17. Let $f(x, y, z) = xy^2z^3 + 3$. Find

a) $f(2, 1, 2)$

$$f(2,1,2) = 2(1)^2(2)^3 + 3 = 2(8) + 3 = 19$$

b) $f(-3, 2, 1)$

$$f(-3, 2, 1) = (-3)(2)^2(1)^3 + 3 = -9$$

c) $f(0, 0, 0)$

$$f(0,0,0) = (0)(0)^2(0)^3 + 3 = 3$$

d) $f(a, a, a)$

$$f(a,a,a) = a(a)^2(a)^3 + 3 = a^6 + 3$$

e) $f(t, t^2, -t)$

$$f(t, t^2, -t) = t(t^2)^2(-t)^3 + 3 = t^5(-t^3) + 3 = -t^8 + 3$$

f) $f(a+b, a-b, b)$

$$f(a+b, a-b, b) = (a+b)(a-b)^2(b)^3 + 3 = (a+b)(a-b)^2b^3 + 3$$

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1-6 Use limits laws and continuity properties to evaluate the limit.

1. $\lim_{(x,y)\to(1,3)} (4xy^2 - x)$

sol:

$= 4(1)(3)^2 - 1$

$= 4(9) - 1$

$= 35$

6) $\lim_{(x,y)\to(4,-2)} x\sqrt[3]{y^3+2x}$

sol:

$= 4\sqrt[3]{(-2)^3 + 2(4)}$

$= 4\sqrt[3]{-8+8}$

$= 4(0)$

$= 0$

2. $\lim_{(x,y)\to(0,0)} \frac{4x-y}{\sin y - 1}$

sol:

$= \frac{4(0)-0}{\sin(0)-1} = \frac{0}{-1} = 0$

3. $\lim_{(x,y)\to(-1,2)} \frac{xy^3}{x+y}$

sol:

$= \frac{(-1)(2)^3}{-1+2} = \frac{-8}{1} = -8$

4. $\lim_{(x,y)\to(1,-3)} e^{2x-y^2}$

sol:

$= e^{2\cdot 1 - (-3)^2} = e^{-7}$

5. $\lim_{(x,y)\to(0,0)} \ln(1+x^2y^3)$

sol:

$= \ln(1 + 0^2(0)^3) = \ln(1+0)$

$= \ln(1) = 0.$

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-8 Show that the limit does not exist by considering the limits as $(x,y) \to (0,0)$ along the coordinate axes.

7.a) $\lim_{(x,y)\to(0,0)} \frac{3}{x^2+2y^2}$

Solve

Let $y = mx$

$= \lim_{m\to 0} \frac{3}{x^2+2(mx)^2}$

$= \frac{3}{x^2}$ limit does not exist.

b) $\lim_{(x,y)\to(0,0)} \frac{x+y}{2x^2+y^2}$

Solve Let $y = mx$

$= \lim_{m\to 0} \frac{x+mx}{2x^2+m^2x^2}$

$= \lim_{m\to 0} \frac{x(1+m)}{x^2(2+m^2)}$

$= \frac{1}{2x}$ limit does not exist.

8.a) $\lim_{(x,y)\to(0,0)} \frac{x-y}{x^2+y^2}$

Solve Let $y = mx$

$= \lim_{m\to 0} \frac{x-mx}{x^2+m^2x^2} = \lim_{m\to 0} \frac{x(1-m)}{x^2(1+m^2)}$

$= \frac{1}{x}$ limit does not exist.

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b) $\lim_{(x,y)\to(0,0)} \frac{\cos xy}{x^2+y^2}$

Solve Let $y = mx$

$= \lim_{m\to 0} \frac{\cos(mx^2)}{x^2+m^2x^2}$

$= \frac{\cos(0)}{x^2} = \frac{1}{x^2}$, limit does not exist.

15. $\lim_{(x,y)\to(0,0)} \frac{xy}{3x^2+2y^2}$

Solve Let $y = mx$

$= \lim_{m\to 0} \frac{x(mx)}{3x^2+2m^2x^2}$

$= \lim_{m\to 0} \frac{mx^2}{x^2(3+2m^2)} = \frac{0}{3+0} = 0$

Limit does not exist.

13. $\lim_{(x,y)\to(0,0)} \frac{x^4-y^4}{x^2+y^2}$

Solve Let $y = mx$

$= \lim_{m\to 0} \frac{x^4-m^4x^4}{x^2+m^2x^2}$

$= \lim_{m\to 0} \frac{x^4(1-m^4)}{x^2(1+m^2)}$

$= x^2 = 0$

Limit exist.

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14. $\lim_{(x,y)\to(0,0)} \dfrac{x^4 - 16y^4}{x^2 + 4y^2}$

Solve

Let $y = mx$

$$= \lim_{x\to 0} \frac{x^4 - 16m^4x^4}{x^2 + 4m^2x^2}$$

$$= \lim_{m\to 0} \frac{x^2(1 - 16m^4)}{x^2(1 + 4m^2)}$$

$$= x^2 \quad = 0^2 = 0$$

Limit exist.

16. $\lim_{(x,y)\to(0,0)} \dfrac{1 - x^2 - y^2}{x^2 + y^2}$

Solve

Let $y = mx$

$$= \lim_{m\to 0} \frac{1 - x^2 - mx^2}{x^2 + m^2x^2}$$

Let $x^2 + y^2 = m$

$$= \lim_{m\to 0} \frac{1 - m}{m}$$

$$= \frac{1-0}{0} = \frac{0}{0} = \infty$$

So the limit does not exist.

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Ex # 13.3

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1. Let $f(x,y) = 3x^3y^2$. Find

a) $f_x(x,y)$

solve

$$\frac{\partial f}{\partial x} = 9x^2y^2$$

b) $f_y(x,y)$

solve

$$\frac{\partial f}{\partial y} = 6x^3y$$

c) $f_x(1,y)$

solve

$$f_x(1,y) = 9(1)^2y^2 = 9y^2$$

d) $f_x(x,1)$

solve

$$f_x(x,1) = 9x^2(1)^2 = 9x^2$$

e) $f_y(1,y)$

solve

$$f_y(1,y) = 6(1)^3y = 6y$$

f) $f_y(x,1)$

solve

$$f_y(x,1) = 6x^3(1) = 6x^3$$

g) $f_x(1,2)$

solve

$$f_x(1,2) = 9(1)^2(2)^2 = 9(4) = 36$$

h) $f_y(1,2)$

$$f_y(1,2) = 6(1)^3(2) = 12$$

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2. Let $z = e^{2x} \sin y$. Find

a) $\frac{\partial z}{\partial x}$

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(e^{2x} \sin y) = 2e^{2x} \sin y$$

b) $\frac{\partial z}{\partial y}$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(e^{2x} \sin y) = e^{2x} \cos y$$

c) $\partial z / \partial x \big|_{(0,y)}$

$$\frac{\partial z}{\partial x}(0,y) = 2e^{2(0)} \sin y = 2 \sin y$$

d) $\partial z / \partial x \big|_{(x,0)}$

$$\frac{\partial z}{\partial x}(x,0) = 2e^{2x} \sin(0) = 0$$

e) $\partial z / \partial y \big|_{(0,y)}$

$$\frac{\partial z}{\partial y}(0,y) = e^{2(0)} \cos y = \cos y$$

f) $\partial z / \partial y \big|_{(x,0)}$

$$\frac{\partial z}{\partial y}(x,0) = e^{2x} \cos(0) = e^{2x}$$

g) $\partial z / \partial x \big|_{(\ln 2, 0)}$

$$\frac{\partial z}{\partial x}(\ln 2, 0) = 2e^{2(\ln 2)} \sin(0) = 0$$

h) $\partial z / \partial y \big|_{(\ln 2, 0)}$

$$\frac{\partial z}{\partial y}(\ln 2, 0) = e^{2(\ln 2)} \cos(0) = 4$$

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3-10 Evaluate the following indicated partial derivatives.

3. $z = 9x^2y - 3x^5y$; $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(9x^2y - 3x^5y)$$
$$= 18xy - 15x^4y$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(9x^2y - 3x^5y)$$
$$= 9x^2 - 3x^5$$

4. $f(x,y) = 10x^2y^4 - 6xy^2 + 10x^2$; $f_x(x,y)$, $f_y(x,y)$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(10x^2y^4 - 6xy^2 + 10x^2)$$
$$= 20xy^4 - 6y^2 + 20x$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(10x^2y^4 - 6xy^2 + 10x^2)$$
$$= 40x^2y^3 - 12xy$$

5. $z = (x^2 + 5x - 2y)^8$; $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(x^2 + 5x - 2y)^8$$
$$= 8(x^2 + 5x - 2y)^8$$
$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(x^2 + 5x - 2y)^8$$
$$= 8(x^2 + 5x - 2y)^7(-2)$$
$$= -16(x^2 + 5x - 2y)^7$$

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6. $f(x,y) = \dfrac{1}{xy^2 - x^2y}$, $f_x(x,y)$, $f_y(x,y)$

solve

$f(x,y) = \dfrac{1}{D}$, where $D = xy^2 - x^2y$

Quotient rule :-

$$\frac{d}{dx}\left(\frac{1}{D}\right) = -\frac{D'}{D^2}$$

$$\frac{\partial D}{\partial x} = y^2 - 2xy$$

so,

$$f_x = -\frac{(y^2 - 2xy)}{(xy^2 - x^2y)^2}$$

$$\frac{\partial D}{\partial y} = 2xy - x^2$$

So,

$$f_y = -\frac{2xy - x^2}{(xy^2 - x^2y)^2}$$

7. $\dfrac{\partial}{\partial p}\left(e^{-7p/q}\right)$, $\dfrac{\partial}{\partial q}\left(e^{-7p/q}\right)$

solve

$$\frac{\partial}{\partial p}\left(e^{-7p/q}\right) = -\frac{7}{q}e^{-7p/q} \qquad \frac{\partial}{\partial q}\left(e^{-7p/q}\right) = \frac{7p}{q^2}e^{-7p/q}$$

8. $\dfrac{\partial}{\partial x}\left(xe^{\sqrt{15xy}}\right)$, $\dfrac{\partial}{\partial y}\left(xe^{\sqrt{15xy}}\right)$

solve

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9. $z = \sin(5x^3y + 7xy^2)$

solve

$$\frac{\partial z}{\partial x} = \cos(5x^3y + 7xy^2)(15x^2y + 7y^2)$$

$$\frac{\partial z}{\partial y} = \cos(5x^3y + 7xy^2)(5x^3 + 14xy)$$

10. $f(x,y) = \cos(2xy^2 - 3x^2y^2)$; $f_x(x,y)$, $f_y(x,y)$.

solve

$$\frac{\partial z}{\partial x} = -(2y^2 - 6xy^2)\sin(2xy^2 - 3x^2y^2).$$

$$\frac{\partial z}{\partial y} = -(4xy - 6x^2y)\sin(2xy^2 - 3x^2y^2).$$

25. $z = 4e^{x^2y^3}$

solve

$$\frac{\partial z}{\partial x} = 4e^{x^2y^3}(2xy^3) = 8xy^3e^{x^2y^3}$$

$$\frac{\partial z}{\partial y} = 4e^{x^2y^3}(3x^2y^2) = 12x^2y^2e^{x^2y^3}$$

26. $z = \cos(x^5y^4)$

solve

$$\frac{\partial z}{\partial x} = -\sin(x^5y^4).(5x^4y^4) = -5x^4y^4\sin(x^5y^4)$$

$$\frac{\partial z}{\partial y} = -\sin(x^5y^4).(4x^5y^3) = -4x^5y^3\sin(x^5y^4)$$

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27. $f = x^3 \ln(1 + xy^{-3/5})$

solve

$$\frac{\partial f}{\partial x} = \left(\frac{\partial}{\partial x}(x^3)\right)\ln(1+xy^{-3/5}) + x^3\left(\frac{\partial}{\partial x}\ln(1+xy^{-3/5})\right)$$

$$= 3x^2 \ln(1+xy^{-3/5}) + x^3\left(\frac{1}{1+xy^{-3/5}} \cdot y^{-3/5}\right)$$

$$= \frac{x^3}{(y^{3/5}+x)} + 3x^2 \ln(1+xy^{-3/5})$$

$$\frac{\partial f}{\partial y} = x^3 \cdot \frac{\partial}{\partial x}\ln(1+xy^{-3/5})$$

$$= x^3 \cdot \left(\frac{1}{1+xy^{-3/5}}\right)\left(-\frac{3}{5}xy^{-8/5}\right)$$

$$= \frac{-3/5\, x^4 y^{-8/5}}{1 + xy^{-3/5}}$$

$$= \frac{-(3/5)\, x^4}{y^{8/5} + xy}$$

28. $f = e^{xy} \sin 4y^2$

solve

$$\frac{\partial f}{\partial x} = \sin(4y^2) \cdot \frac{\partial}{\partial x}(e^{xy})$$

$$= \sin(4y^2) \cdot (e^{xy} \cdot y)$$

$$= y e^{xy} \sin(4y^2)$$

$$\frac{\partial f}{\partial y} = \left(\frac{\partial}{\partial y}(e^{xy})\right)\sin(4y^2) + e^{xy}\left(\frac{\partial}{\partial y}\sin(4y^2)\right)$$

$$= (x e^{xy})\sin(4y^2) + e^{xy}(\cos(4y^2) \cdot 8y)$$

$$= e^{xy}(x \sin(4y^2) + 8y \cos(4y^2))$$

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29. $z = \dfrac{xy}{x^2+y^2}$

Solu

$$\frac{\partial z}{\partial x} = \frac{\left(\frac{\partial}{\partial x}(xy)\right)(x^2+y^2) - (xy)\left(\frac{\partial}{\partial x}(x^2+y^2)\right)}{(x^2+y^2)^2}$$

$$= \frac{y(x^2+y^2) - xy(2x)}{(x^2+y^2)^2} = \frac{x^2y + y^3 - 2x^2y}{(x^2+y^2)^2}$$

$$= \frac{y^3 - x^2y}{(x^2+y^2)^2} = -\frac{y(x^2-y^2)}{(x^2+y^2)^2}$$ Ans

$$\frac{\partial z}{\partial y} = \frac{\left(\frac{\partial}{\partial y}(xy)\right)(x^2+y^2) - (xy)\left(\frac{\partial}{\partial y}(x^2+y^2)\right)}{(x^2+y^2)^2}$$

$$= \frac{x(x^2+y^2) - xy(2y)}{(x^2+y^2)^2} = \frac{x^3 + xy^2 - 2xy^2}{(x^2+y^2)^2}$$

$$= \frac{x^3 - xy^2}{(x^2+y^2)^2} = \frac{x(x^2-y^2)}{(x^2+y^2)^2}$$ Ans

30. $z = \dfrac{x^2y^3}{\sqrt{x+y}}$

Solu

$$\frac{\partial z}{\partial x} = \frac{(2xy^3)(\sqrt{x+y}) - (x^2y^3)\left(\frac{1}{2\sqrt{x+y}}\right)}{(\sqrt{x+y})^2}$$

× and ÷ by $2\sqrt{x+y}$ to simplify.

$$= \frac{2xy^3(2(x+y)) - x^2y^3}{2(x+y)\sqrt{x+y}} = \frac{4x^2y^3 + 4xy^4 - x^2y^3}{2(x+y)^{3/2}}$$

$$= \frac{3x^2y^3 + 4xy^4}{2(x+y)^{3/2}} = \frac{xy^3(3x+4y)}{2(x+y)^{3/2}}$$ Ans

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$$\frac{\partial z}{\partial y} = \frac{(3x^2y^2)(\sqrt{x+y}) - (x^2y^3)\left(\frac{1}{2\sqrt{x+y}}\right)}{(\sqrt{x+y})^2}$$

× and ÷ by $2\sqrt{x+y}$ to simplify.

$$= \frac{3x^2y^2(2(x+y)) - x^2y^3}{2(x+y)\sqrt{x+y}} = \frac{6x^3y^2 + 6x^2y^3 - x^2y^3}{2(x+y)^{3/2}}$$

$$= \frac{6x^3y^2 + 5x^2y^3}{2(x+y)^{3/2}} = \frac{x^2y^2(6x+5y)}{2(x+y)^{3/2}}$$ thm

31. $f(x,y) = \sqrt{3x^5y - 7x^3y}$

$$f_x(x,y) = \frac{1}{2}(3x^5y - 7x^3y)^{-\frac{1}{2}} \cdot (15x^4y - 21x^2y)$$

$$= \left(\frac{3}{2}\right)x^2y(5x^2 - 7) \cdot (3x^5 - 7x^3y)^{-\frac{1}{2}}$$

$$f_y(x,y) = \frac{1}{2}(3x^5y - 7x^3y)^{-\frac{1}{2}} \cdot (3x^5 - 7x^3)$$

$$= \left(\frac{1}{2}\right)x^3(3x^2 - 7)(3x^5y - 7x^3y)^{-\frac{1}{2}}$$ thm

32. $f(x,y) = \frac{x+y}{x-y}$

$$f_x(x,y) = \frac{(1)(x-y) - (x+y)(1)}{(x-y)^2} = \frac{x-y-x-y}{(x-y)^2}$$

$$= \frac{-2y}{(x-y)^2}$$ thm

$$f_y(x,y) = \frac{(1)(x-y) - (x+y)(-1)}{(x-y)^2} = \frac{x-y+x+y}{(x-y)^2}$$

$$= \frac{2x}{(x-y)^2}$$ thm

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37. $f(x,y) = 9 - x^2 - 7y^3$; $f_x(3,1)$, $f_y(3,1)$

solution

$f_x(x,y) = -2x$, $f_x(3,1) = \boxed{-6}$

$f_y(x,y) = -21y^2$, $f_y(3,1) = \boxed{-21}$

38. $f(x,y) = x^2 y e^{xy}$; $\partial f/\partial x\,(1,1)$, $\partial f/\partial y\,(1,1)$.

solution

$$\frac{\partial f}{\partial x} = (2xy)(e^{xy}) + (x^2y)(ye^{xy})$$

$$= 2xye^{xy} + x^2y^2e^{xy}$$

$$\frac{\partial f}{\partial x}(1,1) = 2(1)(1)e^{(1)(1)} + (1)^2(1)^2e^{(1)(1)} = 2e + e = 3e$$

$$\frac{\partial f}{\partial y} = (x^2)(e^{xy}) + (x^2y)(xe^{xy}) = x^2e^{xy} + x^3ye^{xy}$$

$$\frac{\partial f}{\partial y}(1,1) = (1)^2e^{(1)(1)} + (1)^3(1)e^{(1)(1)} = e + e = 2e$$ ans

39. $z = \sqrt{x^2 + 4y^2}$; $\partial z/\partial x\,(1,2)$, $\partial z/\partial y\,(1,2)$

solution

$$\frac{\partial z}{\partial x} = \frac{1}{2\sqrt{x^2+4y^2}} \cdot (2x) = \frac{x}{\sqrt{x^2+4y^2}}$$

$$\frac{\partial z}{\partial x}(1,2) = \frac{1}{\sqrt{(1)^2+4(2)^2}} = \frac{1}{\sqrt{1+16}} = \frac{1}{\sqrt{17}}$$

$$\frac{\partial z}{\partial y} = \frac{1}{2\sqrt{x^2+4y^2}} \cdot (8y) = \frac{4y}{\sqrt{x^2+4y^2}}$$

$$\frac{\partial z}{\partial y}(1,2) = \frac{4(2)}{\sqrt{1^2+4(2)^2}} = \frac{8}{\sqrt{1+16}} = \frac{8}{\sqrt{17}}$$ ans

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0. $w = x^2 \cos xy$; $\partial w/\partial x\,(1/2, \pi)$, $\partial w/\partial y\,(1/2, \pi)$

solve

$$\frac{\partial w}{\partial x} = (2x)(\cos(xy)) + (-\sin(xy))(x^2)$$

$$= 2x\cos(xy) - x^2 y \sin(xy)$$

$$\frac{\partial w}{\partial x}\left(\frac{1}{2}, \pi\right) = 2\left(\frac{1}{2}\right)\cos\left(\frac{\pi}{2}\right) - \left(\frac{1}{2}\right)^2(\pi)\sin\left(\frac{\pi}{2}\right)$$

$$= 1(0) - \frac{\pi}{4}(1) = -\frac{\pi}{4}$$ Ans

$$\frac{\partial w}{\partial y} = x^2(-\sin(xy)\cdot x) = -x^3\sin(xy)$$

$$\frac{\partial w}{\partial y}\left(\frac{1}{2}, \pi\right) = -\left(\frac{1}{2}\right)^3 \sin\left(\frac{\pi}{2}\right) = -\frac{1}{8}(1) = -\frac{1}{8}$$ Ans

41. Let $f(x, y, z) = x^2 y^4 z^3 + xy + z^2 + 1$. Find

a) $f_x(x, y, z)$

solve

$$\frac{\partial f}{\partial x} = 2xy^4z^3 + y$$ Ans

b) $f_y(x, y, z)$

solve

$$\frac{\partial f}{\partial y} = 4x^2y^3z^3 + x$$ Ans

c) $f_z(x, y, z)$

solve

$$\frac{\partial f}{\partial z} = 3x^2y^4z^2 + 2z$$ Ans

d) $f_x(1, y, z)$

solve

$$\frac{\partial f}{\partial x}(1, y, z) = 2y^4z^3 + y$$ Ans

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e) $f_y(1,2,z)$

solve

$\frac{\partial f}{\partial y}(1,2,z) = 32z^3 + 1$

f) $f_z(1,2,3)$

solve

$\frac{\partial f}{\partial z}(1,2,3) = 438$

42. Let $w = x^2 y \cos z$. Find

a) $\partial w/\partial x\,(x,y,z)$

solve

$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x}(x^2 y \cos z) = 2xy \cos z$

b) $\partial w/\partial y\,(x,y,z)$

solve

$\frac{\partial w}{\partial y} = \frac{\partial}{\partial y}(x^2 y \cos z) = x^2 \cos z$

c) $\partial w/\partial z\,(x,y,z)$

solve

$\frac{\partial w}{\partial z} = \frac{\partial}{\partial z}(x^2 y \cos z) = -x^2 y \sin z$

d) $\partial w/\partial x\,(2,y,z)$

solve

$\frac{\partial w}{\partial x}(2,y,z) = 4y \cos z$

e) $\partial w/\partial y\,(2,1,z)$

$\frac{\partial w}{\partial y}(2,1,z) = 4 \cos z$

f) $\partial w/\partial z\,(2,1,0)$

$\frac{\partial w}{\partial z}(2,1,0) = 0$

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1. Let $z = \sqrt{x}\cos y$. Find

a) $\partial^2 z / \partial x^2$

sol:

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(\sqrt{x}\cos y) = \frac{\cos y}{2\sqrt{x}}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\cos y}{2\sqrt{x}}\right) = \frac{\partial}{\partial x}\left(\frac{1}{2}x^{-\frac{1}{2}}\cos y\right)$$

$$= \frac{1}{2}\cos y \cdot \left(-\frac{1}{2}x^{-3/2}\right) = -\frac{\cos y}{4\sqrt{x^3}}$$ Ans

b) $\partial^2 z / \partial y^2$

sol:

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(\sqrt{x}\cos y) = -\sqrt{x}\sin y$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}(-\sqrt{x}\sin y) = -\sqrt{x}\cos y$$ Ans

c) $\partial^2 z / \partial x \partial y$

sol: We differentiate $\partial z/\partial y$ with respect to x.

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}(-\sqrt{x}\sin y) = \frac{\partial}{\partial x}(-x^{\frac{1}{2}}\sin y)$$

$$= -\sin y \cdot \left(\frac{1}{2}x^{-1/2}\right) = -\frac{\sin y}{2\sqrt{x}}$$ Ans

d) $\partial^2 z / \partial y \partial x$

sol: We differentiate $\partial z/\partial x$ with respect to y.

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial x}\left(\frac{\cos y}{2\sqrt{x}}\right) = \frac{1}{2\sqrt{x}} \cdot (-\sin y) = -\frac{\sin y}{2\sqrt{x}}$$ Ans

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82. Let $f(x,y) = 4x^2 - 2y + 7x^4y^5$. Find

a) f_{xx}

solve

$$f_x = 8x + 28x^3y^5$$

$$f_{xx} = 8 + 84x^2y^5$$ done

b) f_{yy}

solve

$$f_y = -2 + 35x^4y^4$$

$$f_{yy} = 140x^4y^3$$ done

c) f_{xy}

solve

$$f_{xy} = 140x^3y^4$$ done

d) f_{yx}

solve

$$f_{yx} = 140x^3y^4$$ done

83. Let $f(x,y) = \sin(3x^2 + 6y^2)$. Find

a) f_{xx}

solve

$$f_x = \cos(3x^2 + 6y^2) \cdot (6x)$$

$$f_{xx} = \cos(3x^2+6y^2)\,6 + 6x\left(-\sin(3x^2+6y^2)(6x)\right)$$
$$= 6\cos(3x^2+6y^2) - 36x^2\sin(3x^2+6y^2)$$ done

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b) f_{yy}

$f_y = \cos(3x^2 + 6y^2)\,12y$

$f_{yy} = -\sin(3x^2 + 6y^2)(12y)(12y) + \cos(3x^2+6y^2)\,12$ Ans

c) f_{xy}

$f_{xy} = -6x \sin(3x^2 - 6y^2)(12y)$

$= -72xy \sin(3x^2 - 6y^2)$. Ans

d) f_{yx}

$f_{yx} = -12y \sin(3x^2 + 6y^2)(6x)$

$= -72xy \sin(3x^2 - 6y^2)$. Ans

84. Let $f(x,y) = xe^{2y}$. Find

a) f_{xx}

$f_x = e^{2y}$, $f_{xx} = 0$ Ans

b) f_{yy}

$f_y = 2xe^{2y}$, $f_{yy} = 4xe^{2y}$ Ans

c) f_{xy}

$f_{xy} = 2e^{2y}$ Ans

d) f_{yx}

$f_{yx} = 2e^{2y}$ Ans

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85-92 Confirm that the mixed second-order partial derivatives of f are the same.

85. $f(x,y) = 4x^2 - 8xy^4 + 7y^5 - 3$

Solve

$f_x = 8x - 8y^4$, $f_{xy} = -32y^3$

$f_y = -32xy^3 + 35y^4$, $f_{yx} = -32y^3$

86. $f(x,y) = \sqrt{x^2+y^2}$

Solve

$$f_x = \frac{2x}{2\sqrt{x^2+y^2}} = \frac{x}{\sqrt{x^2+y^2}}$$

$$f_{xy} = \frac{-xy}{(x^2+y^2)^{3/2}}$$

$$f_y = \frac{2y}{2\sqrt{x^2+y^2}} = \frac{y}{\sqrt{x^2+y^2}}$$

$$f_{yx} = \frac{-xy}{(x^2+y^2)^{3/2}}$$

87. $f(x,y) = e^x \cos y$

Solve

$f_x = e^x \cos y$, $f_{xy} = -e^x \sin y$

$f_y = -e^x \sin y$, $f_{yx} = -e^x \sin y$

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88. $f(x,y) = e^{x-y^2}$

Solve

$f_x = e^{x-y^2}$, $f_{xy} = -2y\,e^{x-y^2}$

$f_y = -2y\,e^{x-y^2}$, $f_{yx} = -2y\,e^{x-y^2}$

89. $f(x,y) = \ln(4x-5y)$

Solve

$f_x = \dfrac{4}{4x-5y}$, $f_{xy} = \dfrac{20}{(4x-5y)^2}$

$f_y = \dfrac{-5}{4x-5y}$, $f_{yx} = \dfrac{20}{(4x-5y)^2}$

90. $f(x,y) = \ln(x^2+y^2)$

Solve

$f_x = \dfrac{2x}{x^2+y^2}$, $f_{xy} = \dfrac{-4xy}{(x^2+y^2)^2}$

$f_y = \dfrac{2y}{x^2+y^2}$, $f_{yx} = \dfrac{-4xy}{(x^2+y^2)^2}$

95. Given $f(x,y) = x^3y^5 - 2x^2y + x$, find

a) f_{xxy}

Solve

$f_x = \dfrac{\partial}{\partial x}(x^3y^5 - 2x^2y + x) = 3x^2y^5 - 4xy + 1$

$f_{xx} = \dfrac{\partial}{\partial x}(3x^2y^5 - 4xy + 1) = 6xy^5 - 4y$

$f_{xxy} = \dfrac{\partial}{\partial y}(6xy^5 - 4y) = 30xy^4 - 4$ Ans

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b) f_{yxy}

solve

$$f_y = \frac{\partial}{\partial y}(x^3y^5 - 2x^2y + x) = 5x^3y^4 - 2x^2$$

$$f_{yx} = \frac{\partial}{\partial x}(5x^3y^4 - 2x^2) = 15x^2y^4 - 4x$$

$$f_{yxy} = \frac{\partial}{\partial y}(15x^2y^4 - 4x) = 60x^2y^3$$ ans

c) f_{yyy}

solve

$$f_y = 5x^3y^4 - 2x^2$$

$$f_{yy} = \frac{\partial}{\partial y}(5x^3y^4 - 2x^2) = 20x^3y^3$$

$$f_{yyy} = \frac{\partial}{\partial y}(20x^3y^3) = 60x^3y^2$$ ans

96. Given $z = (2x - y)^5$, find

a) $\frac{\partial^3 z}{\partial y \partial x \partial y}$

solve

$$\frac{\partial z}{\partial y} = 5(2x-y)^4 \cdot \frac{\partial}{\partial y}(2x-y) = -5(2x-y)^4$$

$$\frac{\partial^2 z}{\partial x \partial y} = -20(2x-y)^3 \frac{\partial}{\partial x}(2x-y) = -40(2x-y)^3$$

$$\frac{\partial^3 z}{\partial y \partial x \partial y} = -120(2x-y)^2 \frac{\partial}{\partial y}(2x-y) = 120(2x-y)^2$$ ans

b) $\frac{\partial^3 z}{\partial x^2 \partial y}$

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From the previous part, we have:

$$\frac{\partial^2 z}{\partial x \partial y} = -40(2x-y)^3$$

$$\frac{\partial^3 z}{\partial x^2 \partial y} = -120(2x-y)^2 \frac{\partial}{\partial x}(2x-y^2) = -240(2x-y)^2$$ Ans

c) $\frac{\partial^4 z}{\partial x^2 \partial y^2}$

From the previous part, we have:

$$\frac{\partial^3 z}{\partial x^2 \partial y} = -240(2x-y)^2$$

Differentiate with respect to y.

$$\frac{\partial^4 z}{\partial x^2 \partial y^2} = -480(2x-y)\frac{\partial}{\partial y}(2x-y) = 480(2x-y)$$ Ans

97. Given $f(x,y) = y^3 e^{-5x}$, find

a) $f_{xyy}(0,1)$

$$f_x = -5y^3 e^{-5x}, \quad f_{xy} = -15y^2 e^{-5x}$$

$$f_{xyy} = -30y^2 e^{-5x}$$

$$f_{xyy}(0,1) = -30(1)^2 e^{-0(5)} = -30e^0$$

$$= -30$$ Ans

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b) $f_{xxx}(0,1)$

soln

$f_x = -5y^3 e^{-5x}$, $f_{xx} = 25y^3 e^{-5x}$

$f_{xxx} = -125y^3 e^{-5x}$

$f_{xxx}(0,1) = -125(1)^3 e^{-5(0)} = -125e^0 = -125$ Ans

c) $f_{yyxx}(0,1)$

soln

$f_y = 3y^2 e^{-5x}$, $f_{yy} = 6y e^{-5x}$

$f_{yyx} = -30y e^{-5x}$, $f_{yyxx} = 150y e^{-5x}$

$f_{yyxx} = 150(1) e^{5(0)} = 150e^0 = 150$ Ans

98. Given $w = e^y \cos x$, find

a) $\left.\dfrac{\partial^3 w}{\partial y^2 \partial x}\right|_{(\pi/4,0)}$

soln

$\dfrac{\partial w}{\partial x} = \dfrac{\partial}{\partial x}(e^y \cos x) = e^y(-\sin x) = -e^y \sin x$

$\dfrac{\partial^2 w}{\partial y \partial x} = \dfrac{\partial}{\partial y}(-e^y \sin x) = -e^y \sin x$

$\dfrac{\partial^3 w}{\partial y^2 \partial x} = \dfrac{\partial}{\partial y}(-e^y \sin x) = -e^y \sin x$

$\dfrac{\partial^3 w}{\partial y^2 \partial x}(\pi/4, 0) = -e^0 \sin\left(\dfrac{\pi}{4}\right) = -1 \cdot \dfrac{1}{\sqrt{2}} = -\dfrac{1}{\sqrt{2}}$ Ans

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(b) $\left.\dfrac{\partial^3 w}{\partial x^2 \partial y}\right|_{(\pi/4,0)}$

solve

$$\frac{\partial w}{\partial y} = \frac{\partial}{\partial y}(e^y \cos x) = e^y \cos x$$

$$\frac{\partial^2 w}{\partial x \partial y} = \frac{\partial}{\partial x}(e^y \cos x) = -e^y \sin x$$

$$\frac{\partial^3 w}{\partial x^2 \partial y} = \frac{\partial}{\partial x}(-e^y \sin x) = -e^y \cos x$$

$$\frac{\partial^3 w}{\partial x^2 \partial y}(\pi/4, 0) = -e^0 \cos\left(\frac{\pi}{4}\right) = -1 \cdot \frac{1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$$ Ans

99. $f(x,y,z) = x^3 y^5 z^7 + x y^2 + y^3 z$. Find

(a) f_{xy}

solve

$$f_x = \frac{\partial}{\partial x}(x^3 y^5 z^7 + x y^2 + y^3 z) = 3x^2 y^5 z^7 + y^2$$

$$f_{xy} = \frac{\partial}{\partial y}(3x^2 y^5 z^7 + y^2) = 15 x^2 y^4 z^7 + 2y$$ Ans

(b) f_{yz}

solve

$$f_y = \frac{\partial}{\partial y}(x^3 y^5 z^7 + x y^2 + y^3 z) = 5x^3 y^4 z^7 + 2xy + 3y^2 z$$

$$f_{yz} = \frac{\partial}{\partial z}(5x^3 y^4 z^7 + 2xy + 3y^2 z) = 35 x^3 y^4 z^6 + 3y^2$$ Ans

(c) f_{xz}

solve

$$f_{xz} = \frac{\partial}{\partial z}(3x^2 y^5 z^7 + y^2) = 21 x^2 y^5 z^6$$ Ans

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d) f_{zzz}

solve

$$f_{zzz} = \frac{\partial}{\partial z}(7x^3y^5z^6 + y^3) = 42x^3y^5z^5$$

e) f_{zyy}

solve

$$f_{zy} = \frac{\partial}{\partial y}(7x^3y^5z^6 + y^3) = 35x^3y^4z^6 + 3y^2$$

$$f_{zyy} = \frac{\partial}{\partial y}(35x^3y^4z^6 + 3y^2) = 140x^3y^3z^6 + 6y$$

f) f_{xxy}

solve

$$f_{xx} = \frac{\partial}{\partial x}(3x^2y^5z^7 + y^2) = 6xy^5z^7$$

$$f_{xxy} = \frac{\partial}{\partial y}(6xy^5z^7) = 30xy^4z^7$$

g) f_{zyx}

solve

$$f_{zy} = 35x^3y^4z^6 + 3y^2$$

$$f_{zyx} = \frac{\partial}{\partial x}(35x^3y^4z^6 + 3y^2) = 105x^2y^4z^6$$

h) f_{xxyz}

solve

$$f_{xxy} = 30xy^4z^7$$

$$f_{xxyz} = \frac{\partial}{\partial z}(30xy^4z^7) = 210xy^4z^6$$

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10. Let $w = (4x - 3y + 2z)^5$. Find

a) $\dfrac{\partial^2 w}{\partial x \partial z}$

solve

$$\frac{\partial w}{\partial z} = 5(4x-3y+2z)^4 \frac{\partial}{\partial z}(4x-3y+2z) = 10(4x-3y+2z)^4$$

$$\frac{\partial^2 w}{\partial x \partial z} = 10.4(4x-3y+2z)^3 \cdot \frac{\partial}{\partial x}(4x-3y+2z)$$

$$= 160(4x-3y+2z)^3$$ Ans

b) $\dfrac{\partial^3 w}{\partial x \partial y \partial z}$

solve

$$\frac{\partial^2 w}{\partial x \partial z} = 160(4x-3y+2z)^3$$

$$\frac{\partial^3 w}{\partial y \partial x \partial z} = 160.3(4x-3y+2z)^2 \cdot \frac{\partial}{\partial y}(4x-3y+2z)$$

$$= 480(4x-3y+2z)^2 \cdot (-3)$$

$$= -1440(4x-3y+2z)^2$$ Ans

c) $\dfrac{\partial^4 w}{\partial z^2 \partial y \partial x}$

solve

$$\frac{\partial^3 w}{\partial x \partial y \partial z} = -1440(4x-3y+2z)^2$$

$$\frac{\partial^4 w}{\partial z^2 \partial y \partial x} = -1440.2(4x-3y+2z) \frac{\partial}{\partial z}(4x-3y+2z)$$

$$= -2880(4x-3y+2z).(2)$$

$$= -5760(4x-3y+2z)$$ Ans

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